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Tunable bandgap of copper sulfide *via* cation exchange as a p-type transparent electrode

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Owing to the inherent challenges of oxide-based p-type TCOs such as the localized O 2p-derived valence band, deep acceptor levels, and the requirement for thermally demanding crystallization, recent research has increasingly focused on non-oxide halide and chalcogenide systems as alternative wide-bandgap semiconductors for transparent electronics. Here, we report a low-temperature cation-exchange approach for tuning CuS transparent electrodes by incorporating Zn into preformed CuS thin films. Unlike conventional non-oxide transparent electrodes such as carbon-based films, metal nano-wires, or conducting polymers, which lack tunable optoelectronic properties and often suffer from limited mechanical durability, the present Zn:CuS film offers a continuously tunable optical bandgap (1.81–2.67 eV) together with a transmittance of >90% at 550 nm, a sheet resistance of $\sim 150 \Omega \text{ sq}^{-1}$, and a negligible sheet-resistance change over 10^4 bending cycles. Integration of 9% Zn:CuS as a transparent electrode with a CVD-grown MoS₂ channel yielded reliable photodetection under 450 nm illumination at DrainV = 1.0 V, with a photoresponsivity of up to $\sim 3 \text{ A W}^{-1}$ and rise/decay times of $\tau_r = 4.3 \text{ s}$ and $\tau_f = 16.1 \text{ s}$. These results establish Zn:CuS as a solution-processable, bandgap-tunable alternative to conventional non-oxide transparent electrodes for next-generation 2D optoelectronics.

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Introduction

Transparent conducting oxides (TCOs) are wide-bandgap semiconductors that combine optical transparency and electrical conductivity, making them indispensable in displays, solar cells, and transparent sensors.¹ High-performance p-type transparent electrodes are essential for realizing transparent devices and complementary integrated optoelectronic devices, both of which underpin the development of fully transparent electronics.^{2,3} While n-type TCOs such as ITO and FTO exhibit high carrier mobility and simple doping chemistry, developing p-type counterparts remains a major challenge due to the localized O 2p-derived valence band that limits hole transport.^{3–7} Consequently, typical Cu- or Ni-based oxides (CuAlO₂,^{8–10}

CuCrO₂,^{11–13} and NiO¹⁴) exhibit conductivities of only $\sim 0.01\text{--}10 \text{ S cm}^{-1}$ three to four orders of magnitude lower than those of n-type TCOs ($\sim 10^4 \text{ S cm}^{-1}$), despite maintaining large bandgaps.^{3,4,15} Moreover, their synthesis typically requires high-temperature ($\sim 400\text{--}900^\circ\text{C}$) pulsed laser deposition (PLD) or sputtering processes, hindering scalable fabrication and integration onto flexible substrates.^{3,5,15} Therefore, achieving low-temperature, high-mobility p-type Transparent Conductors (TCs) is a key step toward fully transparent electronic platforms.

These intrinsic limitations rooted in the localized O 2p-derived valence band, the deep acceptor levels associated with cation or anion defects, and the high crystallization temperatures required for ordered film formation have motivated a shift toward non-oxide halide and chalcogenide systems, whose more dispersed valence bands and softer lattice chemistries offer more tractable routes to wide-bandgap p-type conduction.^{4,6,15} In particular, sulfur-doped copper(i) iodide (S-CuI) has attracted attention as a model halide system, where controlled sulfur incorporation modulates Cu-vacancy (V_{Cu}) concentration and enhances valence-band dispersion, leading to conductivities exceeding 300 S cm^{-1} , while maintaining high optical transmittance.^{16–22} Beyond halides, Cu–Zn chalcogenide alloys have emerged as another platform for transparent electronics; nanocrystalline (CuS)_x(ZnS)_{1–x} thin films

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exhibit p-type conductivities approaching 10^3 S cm^{-1} and optical transmittance above 70%.^{23,24} Importantly, these non-oxide chemical routes can simultaneously meet the conductivity (10^2 – 10^3 S cm^{-1}) and visible-transmittance (>80%) benchmarks outlined above through low-temperature, solution-compatible synthesis, suggesting a viable path toward scalable non-oxide p-type transparent electrodes.^{3,4}

To realize Zn-incorporated CuS (Zn:CuS) thin films with desirable optoelectronic properties, various synthesis routes have been explored, including chemical and physical deposition methods.^{23–27} Among them, chemical bath deposition (CBD) has been widely used due to its simplicity and low processing temperature (<100 °C), producing p-type $(\text{CuS})_x(\text{ZnS})_{1-x}$ nanocomposites with conductivities on the order of 10^2 – 10^3 S cm^{-1} .^{23,28} In parallel, techniques such as spray pyrolysis,²⁷ PLD,²⁴ and thermal evaporation²⁶ have also been explored to improve crystallinity and stoichiometric control, since the former reduces grain-boundary scattering and enhances hole mobility, while the latter governs the V_{Cu} density that determines the carrier concentration and phase purity in Cu-based p-type chalcogenides. However, these approaches generally involve elevated temperatures (~300–500 °C) and multiple vacuum-based steps, making the overall fabrication complex. Consequently, the development of low-temperature processes for Zn–Cu–S thin films remains an important direction toward high-performance p-type transparent electrodes.

Here, we introduce a cation-exchange strategy for doping Zn ions into preformed CuS films *via* controlled substitution at Cu sites, while preserving the host lattice. The resulting Zn:CuS electrodes show >90% transmittance at 550 nm, a sheet resistance of $\sim 150 \Omega \text{ sq}^{-1}$, and an optical bandgap widened from 1.81 to 2.67 eV after Zn incorporation *via* cation exchange and preserve these properties with negligible degradation following 10^4 mechanical bending cycles. To demonstrate their functional utility, the films are integrated with a MoS_2 channel to form a transparent photodetector, which shows strong photocurrent and stable on–off cycling. These results establish the capability of solution-processed non-oxide p-type electrodes to support robust optoelectronic performance, underscoring their viability for next-generation transparent and flexible photodetector platforms.

Results and discussion

The fabrication process of Zn:CuS thin films is illustrated in Fig. 1a, which involves a three-step procedure combining vapor-phase synthesis and solution-phase cation exchange. In the first step, a thin Cu metal film is deposited *via* thermal evaporation. This is followed by sulfurization, where the Cu film is exposed to H_2S gas, leading to the formation of a uniform CuS electrode. The mechanism underlying this Cu-to-CuS conversion, including the role of the native oxide intermediate and its subsequent reaction with H_2S , is discussed in detail in SI Note S1 and Fig. S1.^{29–36} To improve the structural

quality of the CuS film, a low-temperature annealing step at 150 °C is subsequently carried out. During the final step, the entire CuS film is dipped into a precisely controlled ZnCl_2 solution. This dipping process induces a cation exchange reaction between Zn ions in the solution and Cu ions within the CuS framework, resulting in the incorporation of Zn into the CuS lattice and the formation of Zn:CuS thin films.

By varying the ZnCl_2 solution concentration from 0.01 M to 1 M on Zn:CuS films deposited on SiO_2 substrates, a systematic increase in Zn incorporation (3% Zn–16% Zn) could be observed by energy-dispersive spectroscopy (EDS) analysis (SI Note S2, Table S1 and Fig. 1b) and further visualized by spatially resolved EDS elemental mapping (Fig. S2), which clearly demonstrates homogeneous incorporation of Zn across the film. Notably, despite the film thickness remaining unchanged across all compositions (SI Note S3 and Fig. S3), the Zn:CuS films exhibit a remarkable enhancement in optical transparency compared to the pristine CuS film, as clearly evidenced by the photographs of the samples on glass substrates (SI Note S4, Fig. S4). This thickness-independent transparency increase provides strong evidence that the enhanced optical transmittance originates from the electronic/optical modification induced by Zn incorporation, thereby confirming the successful incorporation of Zn into the CuS framework.

The favorable cation exchange between Zn^{2+} and Cu^+ in metal sulfide lattices is well established: Zn^{2+} can effectively replace Cu^+ through interstitial or vacancy mediated diffusion, driven by local electronic and structural compatibility. Importantly, this exchange can proceed under mild thermal conditions (≤ 100 °C) because Cu^+ exhibits exceptionally high ionic mobility in the layered covellite framework, the near-identical tetrahedral ionic radii of Cu^+ and Zn^{2+} minimize the activation barrier for site substitution, and the intrinsic Cu deficiency of CuS provides abundant pre-existing vacancies that accommodate incoming Zn^{2+} ions without requiring high-temperature defect generation.³⁷ In Cu_{2-x}S systems, elevated temperatures or ligand mediated environments facilitate this exchange, enabling Zn incorporation to systematically reduce hole concentration by occupying acceptor rich Cu vacancies.³⁸ Furthermore, mechanistic and Density Functional Theory (DFT) studies on similar copper chalcogenides (*e.g.*, Cu_2Se) identify vacancy assisted pathways and Coulomb-stabilized defect formation as key drivers for ion migration and substitution.³⁹

To elucidate the electronic-level modifications induced by Zn incorporation, ultraviolet photoelectron spectroscopy (UPS) measurements were performed (SI Note S5 and Fig. S5a–c). The UPS spectra reveal tunable shifts in both the secondary electron cutoff (E_{cutoff}) and the Fermi level (E_{f}) as the Zn concentration increases from 3% to 16%. These shifts indicate an upward movement of E_{f} accompanied by a reduction in work function, decreasing from 5.51 eV in pristine CuS to 4.90 eV in the 16% Zn:CuS film. Such behavior is consistent with the thermodynamic preference for Zn^{2+} occupying Cu^+ sites, as the two ions possess nearly identical ionic radii (both $\sim 0.60 \text{ \AA}$),⁴⁰ enabling substitution with minimal lattice distortion.⁴¹

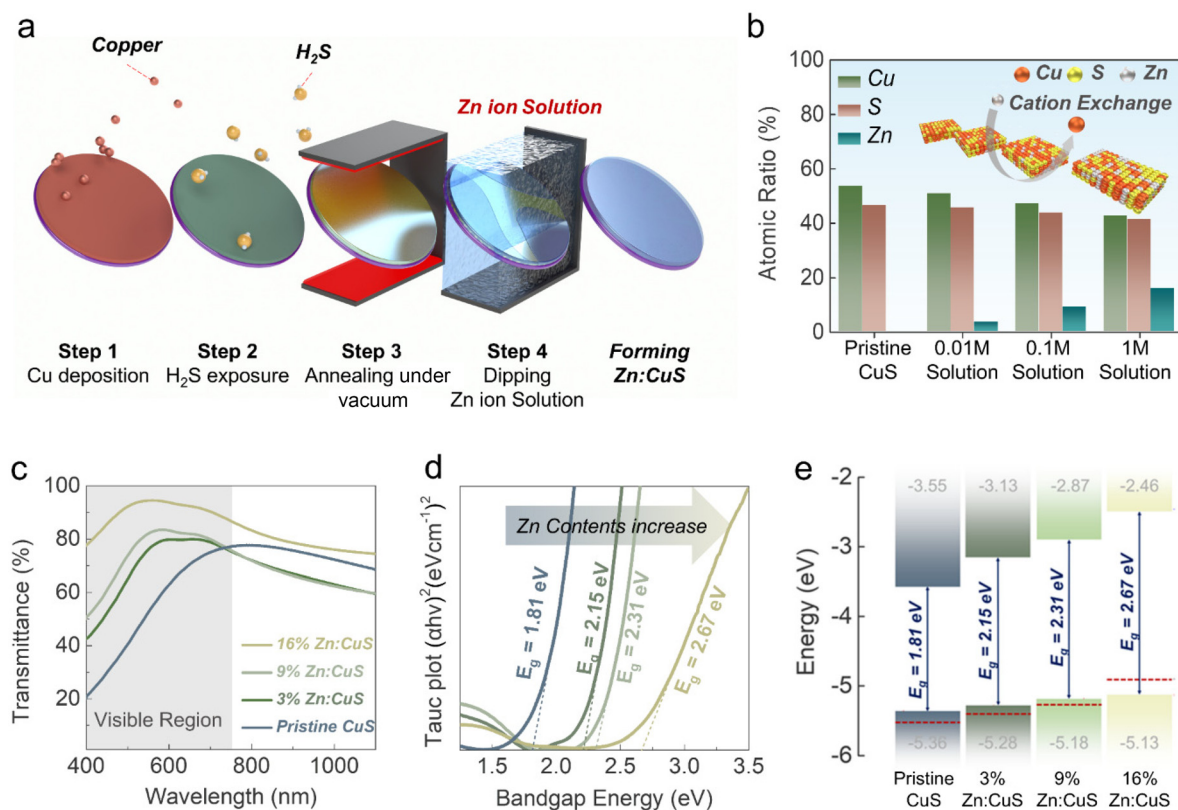


Fig. 1 Modulating CuS band structure and transmittance through Zn ion doping. (a) Schematic illustration of the Zn:CuS film fabrication process. (b) Atomic composition analysis from energy-dispersive X-ray spectroscopy (EDS) confirms the increasing Zn content and decreasing Cu content with higher ZnCl₂ concentrations. (c) Optical transmittance spectra of pristine and Zn-doped CuS films in the 400–1100 nm range, showing enhanced transmittance with increasing Zn content. (d) Tauc plot derived from UV-vis absorption data. (e) Energy band diagrams derived from UPS-measured valence-band maximum positions combined with the Tauc-derived apparent optical bandgap.

First-principles calculations in references further show that the formation energy of Zn substitution at Cu sites is significantly lower than that of interstitial incorporation or substitution on other lattice positions, making Cu-site incorporation the energetically favored pathway.^{38,39,41} This substitution is additionally stabilized by the intrinsic tendency of Cu-based chalcogenides and halides to form Cu vacancies, which provides a natural charge-compensation environment for aliovalent Zn²⁺ doping. This UPS-based band alignment evolution directly supports the interpretation that Zn doping modifies the electronic structure at the band edges, thereby influencing the overall optoelectronic characteristics of the Zn:CuS electrodes.^{38,41–43}

To further examine how these electronic-structure changes manifest in optical behavior, we evaluated the transmittance spectra and optical bandgap of the Zn:CuS films (Fig. 1c and d and SI Note S6). As shown in Fig. 1c, the transmittance across the 400–700 nm visible range steadily increases with Zn content, indicating enhanced optical transparency. This trend aligns with the widening of the apparent optical bandgap extracted from the Tauc analysis, which expands from 1.81 eV in pristine CuS to 2.67 eV at 16% Zn incorporation.^{23,24} Shifts in the valence-band and Fermi-level positions and the

widening of the optical bandgap (Fig. 1c and d) are both observed to trend consistently with increasing Zn content (SI Note S5), consistent with Zn substitution suppressing Cu-vacancy-derived hole density.

To enable direct comparison with prior Cu–Zn–S and related transparent-conductor literature, which predominantly reports Tauc-derived optical bandgaps, we adopted the conventional Tauc analysis as our primary benchmarking metric throughout the manuscript. To additionally obtain a more rigorous, operator-independent estimate of the genuine optical band-edge shift free from the influence of a manually selected linear fitting region, we applied the unified Boltzmann-sigmoid methodology of Zanatta (Zanatta plot) to the same transmittance spectra (SI Note S6).⁴⁴ This analysis yields a genuine band-edge widening of 0.71 eV (2.00 to 2.71 eV), compared to the 0.86 eV apparent shift obtained *via* the Tauc method, with the ~0.15 eV difference attributable to a partial contribution from free-carrier absorption suppression as hole concentration decreases with increasing Zn content.

To investigate the effect of Zn ion doping on the electrical and mechanical performance of CuS thin films, systematic measurements were conducted, including carrier concentration, mobility, sheet resistance, optical transmittance, and

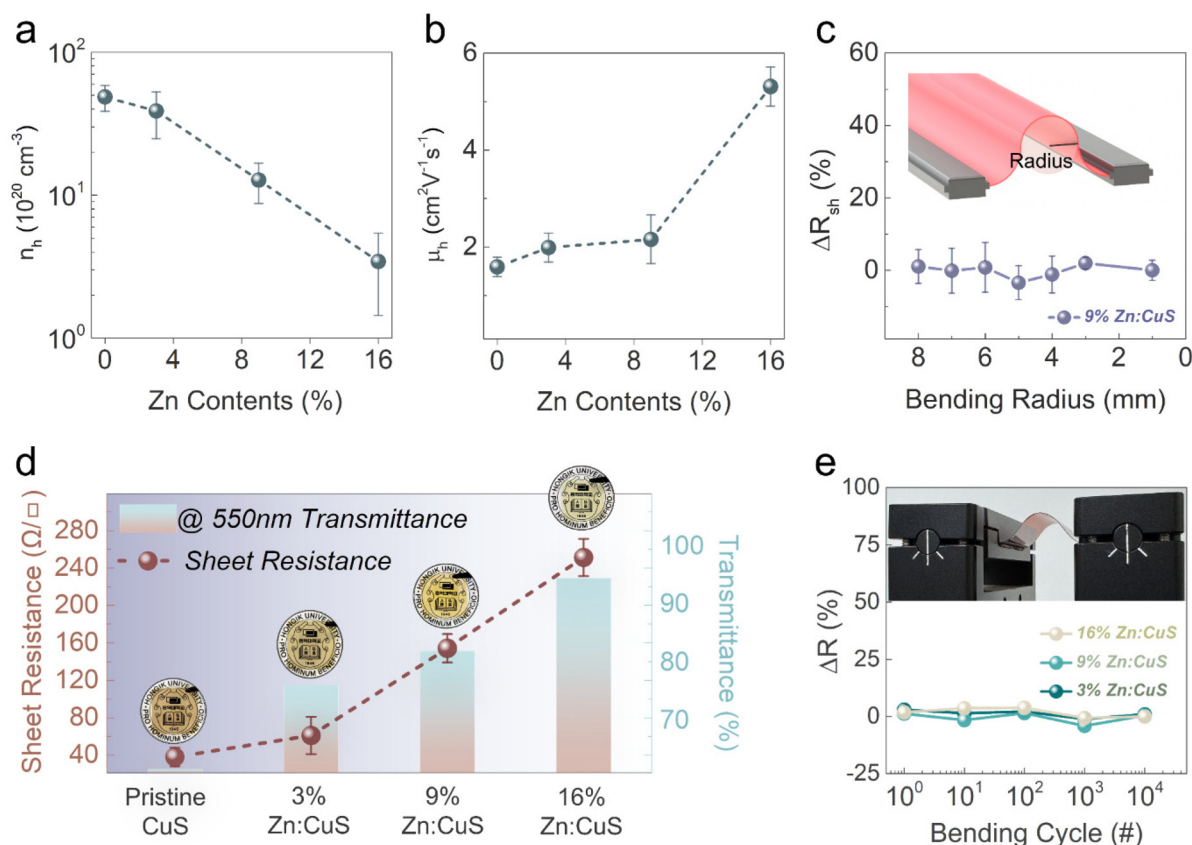


Fig. 2 Electrical and mechanical performance of Zn:CuS thin films. (a) Hole concentration (n_h) as a function of Zn content. (b) Hole mobility (μ_h) extracted from Hall measurements. (c) Sheet resistance as a function of bending radius for 9% Zn:CuS, confirming mechanical stability and minimal change in sheet resistance. (d) Simultaneous evaluation of sheet resistance (left axis) and optical transmittance at 550 nm (right axis) for pristine and Zn-doped CuS films. Increasing Zn doping enhances transmittance while moderately increasing resistance, highlighting a trade-off between optical and electrical performance. (e) Bending fatigue test results showing negligible resistance change (ΔR) over 10^4 cycles for Zn:CuS films.

flexibility. As shown in Fig. 2a, increasing the Zn doping level leads to a pronounced reduction in hole concentration, which aligns with prior reports showing that Zn^{2+} incorporation compensates the acceptor-like Cu vacancies responsible for p-type conduction in the undoped Cu-based system.⁴¹ We note that all compositions, including 16% Zn, retain a positive Hall coefficient and remain p-type throughout the series; the reduction in hole concentration reflects a systematic suppression of native acceptor states rather than a transition in conduction type (SI Note S7).

Conversely, the modest increase in hole mobility with Zn doping (Fig. 2b) can be attributed to Zn^{2+} substituting Cu sites and suppressing Cu-vacancy derived charged defects. This reduction in vacancy-related scattering centers decreases ionized-impurity scattering, enabling more efficient hole transport.^{23,24,41} We note that as the hole concentration approaches its lowest values at 16% Zn, the relative uncertainty in the Hall-derived carrier concentration increases substantially, and the single-carrier assumption underlying standard Hall analysis becomes less robust; accordingly, the mobility value at this composition is interpreted with corresponding caution rather than as a straightforward extension of the

ionized-impurity-scattering trend observed at lower Zn content. Despite the decrease in carrier concentration, the Zn:CuS films maintain a reasonably low sheet resistance, especially at 3% and 9% Zn content, while achieving enhanced optical transmittance at 550 nm (Fig. 2d). Furthermore, to evaluate the mechanical reliability of Zn:CuS films, the 9% Zn:CuS sample, which represents an intermediate doping level that balances electrical conductivity and optical transmittance, was selected as a representative case. As shown in Fig. 2c and e, the 9% Zn:CuS film exhibits excellent mechanical durability, maintaining stable sheet resistance across a range of bending radii (Fig. 2c) and throughout 10^4 bending cycles (Fig. 2e), thereby confirming its potential for use in flexible electronics.

Based on the data shown in Fig. 3, the chemical composition, bonding characteristics, and structural evolution of Zn:CuS thin films were systematically analyzed. X-ray photoelectron spectroscopy (XPS) was employed to investigate the elemental states and chemical environment. As shown in Fig. 3a–c, the Cu 2p and S 2p core level spectra exhibit clear chemical shifts as Zn ion content increases. The Cu 2p_{3/2} and 2p_{1/2} peaks in Fig. 3a show a slight shift toward higher binding energy, suggesting a change in the local bonding

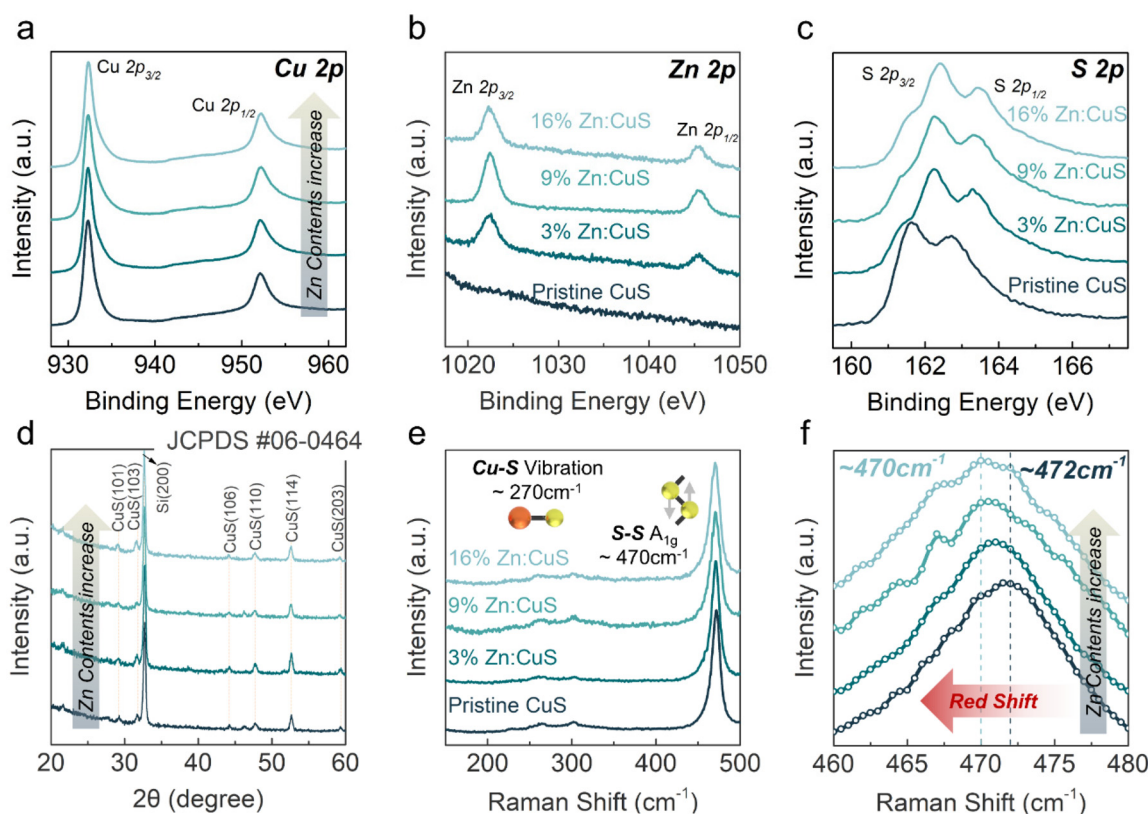


Fig. 3 Structural and bonding analysis of Zn-doped CuS thin films. (a–c) X-ray photoelectron spectroscopy (XPS) spectra of Cu 2p, Zn 2p, and S 2p core levels for pristine and Zn:CuS films. (d) X-ray diffraction (XRD) patterns of pristine CuS and Zn:CuS. Peak positions slightly shift toward lower 2θ values with increasing Zn contents. (e) Raman spectra showing Cu–S ($\sim 270\text{ cm}^{-1}$) and S–S A_{1g} ($\sim 470\text{ cm}^{-1}$) vibrational modes. (f) Magnified view of the S–S A_{1g} mode reveals a gradual red shift from $\sim 472\text{ cm}^{-1}$ to $\sim 470\text{ cm}^{-1}$ with increasing Zn contents.

environment due to partial substitution of Cu^+ by Zn^{2+} . In Fig. 3b, the Zn 2p signal becomes increasingly pronounced as Zn ion content increases, confirming successful incorporation of Zn ions.²⁷ The systematic increase in XPS binding energies observed for the Cu 2p and S 2p core levels with increasing Zn content can be directly interpreted through the relative movement of the Fermi level (SI Note S9 and Fig. S8). In XPS, the binding energy is defined as

$$E_b = h\nu - E_k - \phi$$

where E_b is the core-level binding energy, $h\nu$ is the constant photon energy, E_k is the measured kinetic energy, and ϕ is the work function.⁴⁵ Since this expression references the Fermi level as the zero-energy point, an upward shift of E_f relative to the band edges increases the energy separation between the core states and E_f , which appears experimentally as a shift toward higher binding energies.^{46–51} This behavior indicates that Zn^{2+} incorporation alters the local electronic density by occupying Cu sites and partially suppressing the formation of Cu vacancy defects that serve as the primary acceptor states in pristine CuS.

This interpretation is further supported by UPS measurements, which show a simultaneous increase in the energy difference between the Fermi level and the valence-band maximum $E_f - E_{\text{VBM}}$ and a significant decrease in work function

from $\sim 5.5\text{ eV}$ to $\sim 4.9\text{ eV}$ (Fig. S5). These shifts are characteristic signatures of an electronic landscape in which acceptor concentrations are diminished, causing E_f to rise closer toward mid-gap or even slightly toward the conduction-band side, a trend widely reported in Zn-doped CuI,⁴¹ $\text{Cu}_x\text{Zn}_{1-x}\text{S}$ alloys,²³ and chloride-doped MoS_2 systems.⁴⁶ In this framework, the combination of XPS and UPS shifts provides a coherent thermodynamic and spectroscopic picture: Zn^{2+} substitution reduces Cu-vacancy acceptors, lowers hole density, weakens ionized-impurity scattering, and induces a modest n-type doping tendency.

These trends can be rationalized within the standard Boltzmann approximation for non-degenerate semiconductor statistics, in which the hole concentration is related to the energetic separation between the Fermi level and the valence-band maximum:

$$p_0 \propto \exp\left(-\frac{E_f - E_{\text{VBM}}}{kT}\right)$$

This relation indicates that the hole concentration decreases exponentially as $E_f - E_{\text{VBM}}$ increases, precisely the trend observed by UPS, where $E_f - E_{\text{VBM}}$ increases progressively with Zn content (Fig. S5), consistent with the corresponding decrease in Hall-derived hole concentration

(Table S3). A detailed theoretical treatment of this relationship is provided in SI Note S8.

X-ray diffraction (XRD) patterns in Fig. 3d retain the characteristic covellite reflections while progressively shifting toward lower 2θ angles with increasing Zn content, indicating that Zn^{2+} is substitutionally incorporated into the CuS lattice rather than forming a separate phase. The magnitude of this shift somewhat exceeds what a Vegard-type estimate would predict from the nearly identical tetrahedral ionic radii of Cu^+ and Zn^{2+} (both $\sim 0.6 \text{ \AA}$),^{40,52} and the response is anisotropic: the c -axis-coupled (101) reflection expands by $\sim 1.3\%$ while the in-plane (110) reflection slightly contracts. Such behavior is consistent with the layered structure of covellite, in which alternating CuS_3 trigonal and CuS_4 tetrahedral sublattices are bridged by covalent S–S dimers (S_2^{2-}) along the c -axis.^{33,53,54} Since the nearly identical tetrahedral ionic radii of Cu^+ and Zn^{2+} alone cannot account for the magnitude of the observed shift, the lattice response must originate from an additional perturbation of the anion sublattice. Solution-phase cation exchange is well known to proceed with partial loss of host-lattice sulfur,^{25,39} which in covellite preferentially weakens the bridging S–S dimers that link adjacent CuS_n sublayers along the c -axis; release of this dimer-mediated compressive constraint naturally accounts for both the c -axis expansion and the simultaneous in-plane contraction observed here.⁵⁵

XRD patterns of pristine and Zn:CuS films (Fig. 3d) confirm that the covellite structural framework is retained across all compositions. Reflections were indexed against the covellite reference pattern; we note that the intense reflection near $2\theta \approx 32.8^\circ$ is convolved with a spurious Si(200) substrate reflection arising from multiple diffraction in θ – 2θ geometry, and was therefore excluded from quantitative lattice-parameter analysis. Using the remaining reflections, the (101) d -spacing increases systematically with Zn content ($+0.74\%$ from pristine to 16% Zn), while the (110) reflection, which probes the a -axis exclusively in the hexagonal covellite structure, remains essentially invariant (-0.04%), indicating an anisotropic lattice distortion rather than a uniform expansion. This distortion is consistent with the moderate genuine optical band-edge shift discussed above and does not require substantial restructuring of the electronic band structure.

This structural picture is supported by the accompanying spectroscopic and compositional evidence. The A_{1g} Raman mode shifts systematically from ~ 472 to $\sim 470 \text{ cm}^{-1}$ with increasing Zn content (Fig. 3f), reflecting softening of the S–S dimer bond,^{27,33,35,56} while EDS analysis (Fig. 1b) shows that the S/(Cu + Zn) ratio decreases from ~ 0.87 in pristine CuS to ~ 0.71 in the 16% Zn:CuS sample, a deviation from pure 1 : 1 cation exchange that points to additional perturbation of the anion sublattice. The non-monotonic intensity of a weak feature near $2\theta \approx 47.5^\circ$, which is most pronounced at intermediate (3–9%) Zn loadings and weakens at 16% Zn, in contrast to the monotonic growth expected of a segregated ZnS phase, together with the continuous (rather than discontinuous) evolution of the Raman and XPS features with Zn content, excludes bulk ZnS phase segregation and supports a single-

phase Zn:CuS solid solution. Analogous lower- 2θ shifts have been reported for Zn^{2+} incorporation in other Cu(I)-based hosts such as CuSCN ⁵⁷ and CuI ,⁴¹ as well as for $\text{Cu}_x\text{Zn}_{1-x}\text{S}$ films prepared by SILAR²⁵ and combinatorial sputtering,²⁴ placing our cation-exchange samples firmly within the established structural trend of the Cu–Zn–S system.

Raman spectroscopy further corroborates that Zn doping perturbs the local bonding environment of CuS while preserving the covellite host lattice (Fig. 3e and f). Across all compositions, the Cu–S vibrational mode at $\sim 270 \text{ cm}^{-1}$ remains clearly visible with only minor variation in frequency and linewidth, consistent with earlier assignments of this band to Cu–S lattice vibrations in covellite-type CuS and confirming that the primary CuS framework is retained upon Zn incorporation.^{56,58} In contrast, the intense Raman band around $470\text{--}472 \text{ cm}^{-1}$, which corresponds to the A_{1g} symmetric stretching of covalent S–S dimers (S_2^{2-}) in the covellite structure, exhibits a systematic shift toward lower frequency as the Zn content increases. In our Zn:CuS films, this trend is consistent with Zn^{2+} substituting Cu sites and compensating Cu-vacancy derived holes, effectively reducing the p-type hole concentration.

To place the electrode performance of the Zn:CuS films in the context of the wider transparent-conductor literature, we evaluated the high-resolution Haacke figure of merit, $\text{FOM}_{\text{Haacke-HR}} = T/(R_{\text{sh}})^{(1/n)}$ with $n = 10$,⁵⁹ which retains the transmittance-weighted proportionality of the original Haacke definition while providing improved resolution for comparisons between closely matched films.⁶⁰ Using the transmittance at 550 nm and the four-point-probe sheet resistance, the films yield $\text{FOM}_{\text{Haacke-HR}}$ values of 0.429 (pristine), 0.505 (3% Zn), 0.496 (9% Zn), and 0.544 (16% Zn); the same ranking is obtained with the conventional Haacke expression. Fig. 4

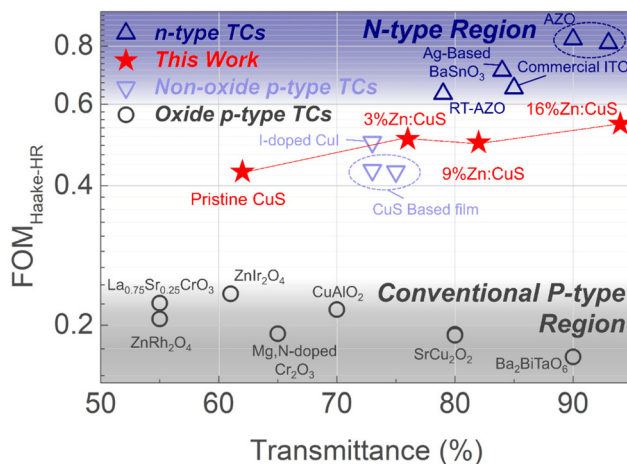


Fig. 4 High-resolution Haacke figure of merit, $\text{FOM}_{\text{Haacke-HR}} = T/(R_{\text{sh}})^{(1/10)}$, plotted against transmittance for the Zn:CuS films of this work (red stars) and representative transparent conductors from the literature: n-type transparent conductors, including commercial ITO (blue triangles), non-oxide p-type transparent conductors (light blue inverted triangles), and oxide p-type transparent conductors (open circles). Shaded regions indicate the ranges typically occupied by n-type and conventional p-type transparent conductors.

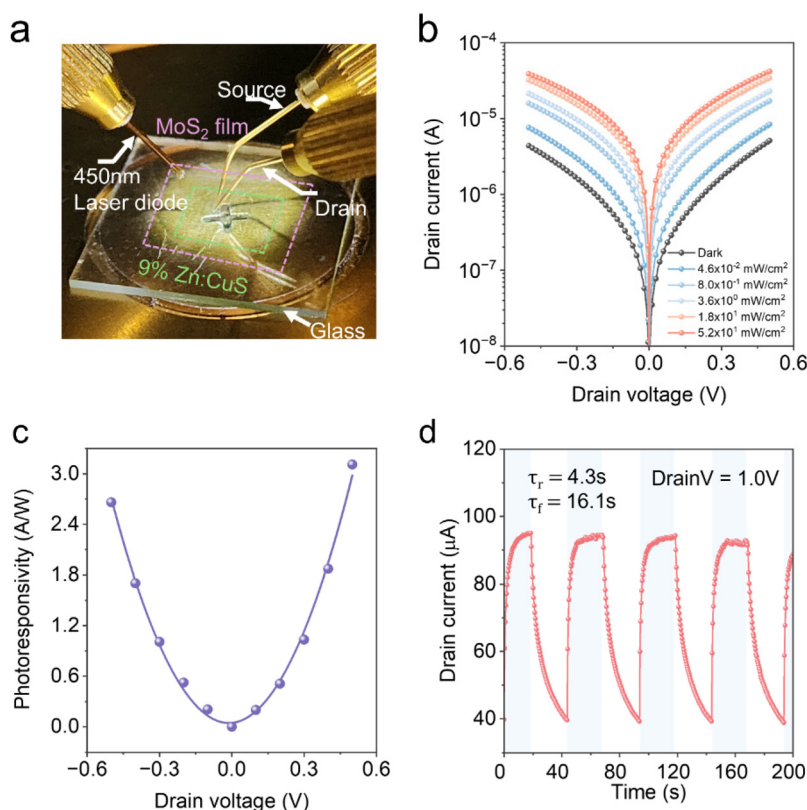


Fig. 5 Transparent MoS₂/Zn:CuS photodetector. (a) Optical image of the two-terminal MoS₂/Zn:CuS transparent device. (b) The output characteristic (I_{DS} – V_{DS}) of the MoS₂/Zn:CuS photodetector at varying illumination intensities. (c) Photoresponsivity as a function of applied drain voltages. (d) Temporal response of photocurrent at a drain voltage of 1.0 V.

compares these values with representative transparent conductors reported in the literature. Conventional oxide p-type transparent conductors cluster below $FOM_{\text{Haake-HR}} \approx 0.25$,^{61–66} reflecting the limited hole conductivity attainable in these systems. Non-oxide p-type conductors, including CuS-based films and I-doped CuI, reach $FOM_{\text{Haake-HR}} \approx 0.42$ – 0.50 ,^{23,67,68} consistent with the greater valence-band dispersion afforded by chalcogenide and halide anions. All four Zn:CuS compositions in this work lie above the oxide p-type group, and the 16% Zn film ($FOM_{\text{Haake-HR}} = 0.544$) narrows the gap toward the n-type region occupied by ITO and doped-oxide multilayers.^{69–72}

To extend the utility of Zn:CuS from transparent conducting films to optoelectronic devices, we integrated the optimized Zn:CuS electrodes with a CVD-grown MoS₂ channel in a lateral two-terminal photodetector geometry. This transparent electrode/2D semiconductor configuration demonstrates the feasibility of constructing fully transparent optoelectronic devices on glass substrates.⁷³ The fabricated device confirms that Zn:CuS functions as an effective transparent electrode and supports reliable photodetection. The drain–source current *versus* drain–source voltage under varying illumination intensities is presented in Fig. 5b on a logarithmic scale: under both dark and illuminated conditions, the device exhibits symmetric and nearly linear current–voltage characteristics, indicating

favorable carrier injection at the MoS₂/Zn:CuS interface.⁷³ Upon 450 nm laser illumination, the drain current increases systematically with light intensity, and the photoresponsivity extracted as a function of applied drain bias reaches a maximum of approximately 3 A W^{-1} at higher bias voltages (Fig. 5c). The dynamic photoresponse in Fig. 5d further shows that, under periodic on/off switching of the 450 nm laser at DrainV = 1.0 V, the drain current rises and recovers reproducibly over many cycles with characteristic rise and decay times of $\tau_r = 4.3 \text{ s}$ and $\tau_f = 16.1 \text{ s}$, respectively. Together, these results verify that Zn:CuS operates as an electronically active and operationally stable component in a lateral, fully transparent MoS₂/Zn:CuS heterostructure on glass, serving as a building block directly compatible with integrated transparent optoelectronics.

Conclusions

In conclusion, we demonstrated a low-temperature cation-exchange strategy for tuning the band structure and carrier properties of CuS transparent electrodes through Zn incorporation. Zn substitution effectively suppressed Cu-vacancy derived acceptors, reduced hole concentration, shifted the Fermi level upward, lowered the work function, and widened

the optical bandgap from 1.81 to 2.67 eV, resulting in >90% transmittance at 550 nm with a sheet resistance of $\sim 150 \Omega \text{ sq}^{-1}$. The optimized Zn:CuS films also showed excellent mechanical durability over 10^4 bending cycles and preserved the covellite host lattice while modulating local bonding and electronic structure. As a proof of concept, transparent MoS₂/Zn:CuS photodetectors on glass exhibited stable photocurrent switching and a photoresponsivity of up to $\sim 3 \text{ A W}^{-1}$, demonstrating that Zn:CuS is a promising non-oxide transparent electrode platform for next-generation optoelectronics.

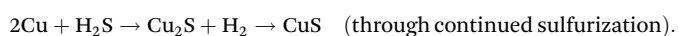
Experimental section

Substrate preparation

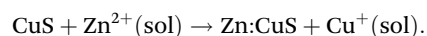
Three types of substrates were employed in this work according to the requirements of each measurement. p⁺-Si substrates with a thermally grown SiO₂ layer were used for all material characterization, including structural (XRD), spectroscopic (XPS, UPS, and EDS), morphological (SEM and AFM), and optical transmittance measurements. Soda-lime glass substrates with Cr 5 nm/Au 30 nm contacts deposited at the four corners were used for Hall-effect measurements in the van der Pauw configuration. PI substrates were used for the mechanical flexibility tests, in which sheet resistance was monitored as a function of bending radius and over repeated bending cycles. All substrates were subjected to an identical cleaning sequence prior to Cu deposition. Following cutting to size, the substrates were blown with N₂ gas to remove particulates and then sequentially sonicated in toluene, acetone, isopropyl alcohol, and deionized water for 10 min in each solvent. The substrates were subsequently rinsed under flowing deionized water and treated with O₂ plasma (O₂ flow rate 50 sccm, 50 W, 5 min) immediately prior to deposition to remove residual organic contamination and improve surface wettability.

Synthesis of the Zn:CuS electrode

The Zn:CuS thin films were synthesized *via* a three-step process consisting of Cu deposition, sulfurization, and subsequent solution based cation exchange. This approach enables controllable Zn incorporation into the CuS framework while maintaining high structural integrity, thereby enhancing both optical and electronic properties. Initially, Cu thin films were deposited onto Si/SiO₂ substrates and flexible substrates (*e.g.*, PI) *via* thermal evaporation using a 99.999% purity Cu metal source (Taewon Scientific Co., South Korea). The deposition was carried out under high vacuum at a controlled rate to ensure uniform thickness and film quality. These Cu films served as the precursor for the formation of copper sulfide. The deposited Cu films were sulfurized by direct exposure to H₂S gas generated from the thermal decomposition of ammonium sulfide solution (Thermo Fisher Scientific, USA). The sulfurization was conducted under ambient pressure at room temperature, enabling the formation of covellite-phase CuS thin films *via* the following reaction:⁷⁴



After sulfurization, an annealing step was performed to improve film crystallinity and phase uniformity. Subsequently, the films were immersed for 10 to 30 min in aqueous ZnCl₂ solutions of varying concentrations (0.01–1 M) to induce cation exchange between Cu⁺ in the CuS lattice and Zn²⁺ ions in solution. This exchange facilitates the incorporation of Zn into the CuS structure without significantly disturbing the covellite phase. The doping mechanism can be represented by:



This solution-phase doping step enables precise control over Zn content, allowing for systematic tuning of the electronic structure and optical properties of the CuS films. The resulting Zn:CuS thin films exhibit modulated band structures and reduced hole concentrations, making them suitable for applications in transparent optoelectronic and van der Waals devices.

Structural and morphological analysis and characterization

The crystallinity and phase composition of pristine and Zn-doped CuS films were analyzed using X-ray diffraction (XRD, Rigaku MiniFlex 600 with Cu K α radiation, $\lambda = 1.5406 \text{ \AA}$, Japan). XRD patterns were obtained to verify the preservation of the covellite CuS phase and to monitor the shift in peak positions with Zn doping. Scanning electron microscopy (SEM, JEOL JSM-IT710HR, Japan) was employed to observe the microstructural evolution, including grain connectivity, film uniformity, and the presence of structural defects. To validate the elemental composition and spatial distribution of Zn within the CuS matrix, energy-dispersive spectroscopy (EDS, Oxford AZtecLive Advanced Ultim Max170, UK) coupled with SEM was used. Elemental mapping of Cu, S, and Zn confirmed the successful and homogeneous incorporation of Zn ions *via* cation exchange, with the covellite phase retained as the dominant structural framework across all compositions, despite a measurable, anisotropic lattice distortion. X-ray photoelectron spectroscopy (XPS, Thermo Scientific K-Alpha+) was used for elemental composition and chemical state analysis, calibrated to the C 1s peak at 284.8 eV. Raman spectra of pristine CuS and Zn:CuS on SiO₂ substrates were obtained using Raman microscopes (SENTERRA II, BRUKER). The measurements were performed in the backscattering configuration with 532 nm excitation from a diode-pumped solid-state laser. The excitation was focused using a 100 $\times$ microscope objective with a numerical aperture of 0.9, resulting in a spot diameter of $\sim 1 \mu\text{m}$, capable of probing single flakes. The scattered signal was sent through an 1800 gr mm^{-1} grating. Spectra were acquired under low laser power (1.5 mW) to prevent sample damage. Typical acquisition times were 3 s per spectrum. All measurements were performed under ambient conditions.

Optical properties

The optical transparency of pristine and Zn:CuS films was evaluated using a UV-vis spectrophotometer (PerkinElmer LAMBDA 365 UV/Vis Spectrophotometer, USA). Transmittance

spectra were recorded in the 400–1100 nm range to investigate the impact of Zn incorporation on transparency and light absorption. To determine the optical bandgap, Tauc plot analysis was performed using the relationship between the absorption coefficient and photon energy, enabling a quantitative assessment of optical bandgap widening induced by Zn doping.

Electrical properties

The electrical transport properties of the Zn:CuS films were investigated using Hall effect measurements in the van der Pauw configuration. These measurements provided carrier concentration, mobility, and conductivity type, revealing a reduction in hole concentration with increasing Zn content and indicating the emergence of n-type doping characteristics. Additionally, four-point probe measurements (Tektronix Keithley 4200A-SCS, USA) were conducted to determine the sheet resistance (R_{sh}) and evaluate the influence of Zn incorporation on charge transport behavior.

Mechanical stability

To evaluate the flexibility and mechanical durability of Zn:CuS films, bending tests were performed using a customized bending test apparatus. Zn:CuS films transferred on PI (polyimide) substrates were subjected to repeated bending cycles with a bending radius of 1 mm (the transfer method is described in SI Note S10). The relative change in resistance (ΔR) was monitored over 10 000 bending cycles to assess the mechanical stability of the films, confirming their potential applicability in flexible optoelectronic and transparent electronic devices.

Photodetector fabrication

The MoS₂/Zn:CuS photodetectors were fabricated by integrating CVD-grown MoS₂ with patterned 9%Zn:CuS transparent electrodes on glass.⁷³ First, a 20 nm-thick Cu film was deposited on a sacrificial substrate and converted into a Zn:CuS layer *via* the cation-exchange process described above. The Zn:CuS film was then patterned into electrode geometries by standard photolithography and wet etching, after which the patterned Zn:CuS electrodes were transferred onto MoS₂/glass substrates using a wet-transfer process.

Conflicts of interest

The authors declare no conflicts of interest.

Data availability

The original contributions presented in this study are included in the article/supplementary information (SI). Supplementary information is available. See DOI: <https://doi.org/10.1039/d6nr02124g>.

Further inquiries can be directed to the corresponding author.

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